

Stoichiometry

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q1 JEE Main 2020 - 2 September (Evening)

The ratio of the mass percentages of 'C' & 'H' and 'C' & 'O' of a saturated acyclic organic compound 'X' are 4: 1 and 3: 4 respectively. Then, the moles of oxygen gas required for complete combustion of two moles of organic compound 'X' is _____.

Q2 JEE Main 2020 - 3 September (Morning)

The mole fraction of glucose ($C_6H_{12}O_6$) in an aqueous binary solution is 0.1. The mass percentage of water in it, to the nearest integer, is

Q3 JEE Main 2020 - 3 September (Evening)

6.023×10^{22} molecules are present in 10g of a substance 'x'. The molarity of a solution containing 5g of substance 'x' in 2L solution is $\times 10^{-3}$

Q4 JEE Main 2020 - 4 September (Morning)

The mass of ammonia in grams produced when 2.8kg of dinitrogen quantitatively reacts with 1 kg of dihydrogen is

Q5 JEE Main 2020 - 4 September (Evening)

A 100 mL solution was made by adding 1.43g of $Na_2CO_3 \cdot xH_2O$. The normality of the solution is 0.1N. The value of x is
(The atomic mass of Na is 23g/mol)

Q6 JEE Main 2020 - 5 September (Morning)

The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is

Q7 JEE Main 2020 - 6 September (Morning)

A solution of two components containing n_1 moles of the 1st component and n_2 , moles of the 2nd component is prepared. M_1 and M_2 are the molecular weights of component 1 and 2 respectively. If d

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is the density of the solution in gmL^{-1} , C_2 is the molarity and x_2 is the mole fraction of the 2nd component, then C_2 can be expressed as

(A) $C_2 = \frac{1000x_2}{M_1 + x_2(M_2 - M_1)}$

(B) $C_2 = \frac{1000dx_2}{M_1 + x_2(M_2 - M_1)}$

(C) $C_2 = \frac{dx_2}{M_2 + x_2(M_2 - M_1)}$

(D) $C_2 = \frac{dx_1}{M_2 + x_2(M_2 - M_1)}$

Q8 JEE Main 2020 - 6 September (Evening)

The average molar mass of chlorine is 35.5 gmol^{-1} . The ratio of ^{35}Cl to ^{37}Cl in naturally occurring chlorine is close to

(A) 01:01

(B) 02:01

(C) 03:01

(D) 04:01

Q9 JEE Main 2020 - 7 January (Evening)

0.6 g of urea on strong heating with NaOH evolves NH_3 . Liberated NH_3 will combine completely with which of the following HCl solution?

(A) 100 mL of 0.2 N HCl

(B) 400 mL of 0.2 N HCl

(C) 100 mL of 0.1 N HCl

(D) 200 mL of 0.2 N HCl

Q10 JEE Main 2020 - 8 January (Morning)

Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10 ppm of iron in 100 kg of wheat is .

Atomic weight : $\text{Fe} = 55.85$; $\text{S} = 32.00$; $\text{O} = 16.00$

Multiply your answer with 100

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Q11 JEE Main 2020 - 8 January (Evening)

$NaClO_3$ is used, even in spacecrafts, to produce O_2 . The daily consumption of pure O_2 by a person is 492 L at 1 atm , 300 K . How much amount of $NaClO_3$, in grams, is required to produce O_2 for the daily consumption of a person at 1 atm , 300 K ?

Q12 JEE Main 2020 - 9 January (Morning)

The mass percentage of nitrogen in histamine is. . . .

Q13 JEE Main 2020 - 9 January (Morning)

The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is. . . .
(Molecular Weight of $HNO_3 = 63$)

Q14 JEE Main 2020 - 9 January (Evening)

10.30 mg of O_2 dissolved into a liter of sea water of density 1.03 g/mL . The concentration of O_2 in ppm is

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Chemistry

Answer Key

Q1 (5)

Q2 (47)

Q3 (25)

Q4 (3400)

Q5 (10)

Q6 (18)

Q7 (B)

Q8 (C)

Q9 (A)

Q10 (496)

Q11 (2130)

Q12 (37.84)

Q13 (14)

Q14 (10)

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Hints & Solutions

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Chemistry

Q1

Let the masses of C, H and O in organic compound X be x , y and z respectively

Mass Moles

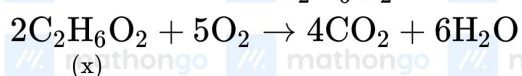
C x 4 12 1

H y 1 3 3

O z $\frac{16}{3}$ 16 1

$\therefore$ Empirical formula : CH_3O

Molecular formula : $\text{C}_2\text{H}_6\text{O}_2$



$\therefore$ Number of moles of O_2 required to oxidise 2 moles of $(X) = 5$

Q2

Mole fraction of glucose in aqueous solution

$$= 0.1$$

Mass percentage of water in it

$$= \left[\frac{0.9 \times 18}{0.9 \times 18 + 0.1 \times 180} \right] \times 100$$

$$= 47.37 \simeq 47$$

Q3

Mass of 6.023×10^{22} molecules of a substance = 10g

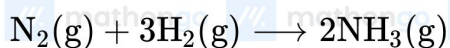
Mass of 6.023×10^{23} molecules of the substance = 100g Molar mass of the substance

$$= 100\text{g mol}^{-1}$$

$$\text{Molarity of the solution} = \frac{5}{100 \times 2} = 2.5 \times 10^{-2}$$

$$= 25 \times 10^{-3}$$

Q4



$$\text{Number of moles of } \text{N}_2 = \frac{2.8 \times 10^3}{28} = 100$$

$$\text{Number of moles of } \text{H}_2 = \frac{1000}{2} = 500$$

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JEE Main 2020 Chapterwise

Chemistry

Number of moles of NH_3 produced = 200

Mass of NH_3 produced = $200 \times 17 = 3400\text{gm}$

Q5

Normality = $\frac{\text{No. of equivalents of solute}}{\text{Volume of solution (in L)}}$

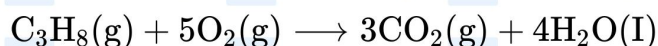
$$0.1 = \frac{1.43}{\frac{(106+18x)}{2} \times 0.1}$$

$$\Rightarrow \frac{106+18x}{2} = 143$$

$$\Rightarrow 18x = 286 - 106 = 180$$

$$x = 10$$

Q6



No. of moles of O_2 required to oxidise 1 mole of propane and 2 moles of butane = $5 + 2 \times \frac{13}{2}$
= 18

Q7

To express C_2 in terms of mole fraction x_2

1st component 2nd component

mole n_1 n_2

m.w M_1

mass n_1M_1 n_2M_2

mass of solution = $n_1M_1 + n_2M_2$

mole fraction $x_2 = \frac{n_2}{n_1+n_2}$

$$n_1 = \frac{n_2(1-x_2)}{x_2}$$

Mass of solution = $n_1M_1 + n_2M_2$

$$= \frac{n_2M_1(1-x_2)}{x_2} + n_2M_2$$

$$= \frac{n_2}{x_2} [M_2x_2 - x_2M_1 + M_1]$$

Volume of solution

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Chemistry

$$= \frac{n_2[M_2x_2 - x_2M_1 + M_1]}{1000dx_2} \text{ Litre}$$

$$C_2 = \frac{1000n_2dx_2}{n_2[M_2x_2 - x_2M_1 + M_1]}$$

$$C_2 = \frac{1000dx_2}{M_1 + x_2(M_2 - M_1)}$$

Q8

$$\text{Average molar mass} = \frac{35 \times 3 + 37 \times 1}{4} = 35.5$$

$$\Rightarrow {}^{35}\text{Cl} : {}^{37}\text{Cl} = 3 : 1$$

Q9

$$2 \times \text{mole of Urea} \equiv \text{mole of } \text{NH}_3 \quad \dots (1)$$

$$\text{mole of } \text{NH}_3 = \text{mole of } \text{HCl} \quad \dots (2)$$

$$\therefore \text{mole of } \text{HCl} = 0.02 \text{ mole}$$

Q10

$$10 = \frac{\text{Mass of } \text{Fe} (\text{in } g)}{100 \times 1000} \times 10^6$$

$$\text{or, mass } \text{Fe} = 1 \text{ g}$$

$$\text{FeSO}_4 \cdot 7\text{H}_2\text{O} (M = 278)$$

$$56 \text{ g in 1 mole}$$

$$1 \text{ g} - \frac{1}{56} \text{ mole} \times \frac{1}{56} \times 278 \text{ g} = 4.96 \text{ g}$$

$$\text{Final ans} = 4.96 \times 100 = 496$$

Q11

\$

$$\text{mol of } \text{NaClO}_3 = \text{mol of } \text{O}_2$$

$$\text{mol of } \text{O}_2 = \frac{PV}{RT} = \frac{1 \times 492}{0.082 \times 300} = 20 \text{ mol}$$

$$\text{mass of } \text{NaClO}_3 = 20 \times 106.5 = 2130 \text{ g}$$

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Chemistry

Q12

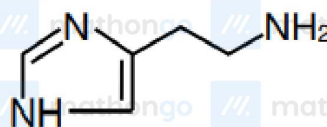
Structure of Histamine is given below

Molecular formula of Histamine is $C_5H_9N_3$

Molecular mass of Histamine is 111

Percentage nitrogen by mass in Histamine

$$= \frac{42}{111} \times 100 = 37.84\%$$



Q13

63% w/w $\rightarrow HNO_3$ solution

$$M = \frac{63 \times 1.4}{63 \times 100} \times 1000 \text{ mol/L}$$

$$M = 14 \text{ mole/L}$$

Q14

$$ppm = \frac{10.3 \times 10^{-3}}{1030} \times 10^6 = 10$$